MATHGR5450, Credit Analytics: HW #1, Due Date: February 20, 2026 at 1:00PM

1. (25 points) We have a random variable X whose distribution function is $F(x)$ and whose density function is $f(x)$.
   (a) (5 points) Prove that $F(X)$ follows uniform distribution over $(0, 1]$.
   (b) (5 points) Take any uniform distribution U over $(0, 1]$, prove that $F^{-1}(U)$ follows the same distribution as X, where $F^{-1}()$ is the inverse function.
   (c) (5 points) Suppose that a random variable Y follows a standard normal distribution whose distribution function is $\Phi(y)$. Show that $\Phi^{-1}(F(X))$ follows the standard normal distribution.
   (d) (10 points) If X is a positive random variable, prove that
   $$E(X) = \int_0^{+\infty} S(x)dx,$$
   where $S(x) = 1-F(x)$ is the survival function, and provide an intuitive explanation for this result.

2. (20 points) Suppose we have a random variable whose hazard rate function is $h(t) = \alpha e^{\beta t}$ where $\alpha > 0$ and $\beta > 0$.
   (a) (10 points) Show that the survival function is
   $$S(t) = e^{\frac{\alpha}{\beta} - \frac{\alpha}{\beta} e^{\beta t}}$$
   (b) (5 points) Derive the distribution function
   (c) (5 points) Derive the density function

3. (25 points) Suppose that we have a constant hazard rate h for a credit risk, and a constant instantaneous forward rate r over the period $[0, T]$.
   (a) (10 points) Suppose a contract which pays \$100 dollar when default occurs before T. Calculate the expected value and the variance of the present value for this contract.
   (b) (10 points) Suppose a contract which pays \$100 dollar per year if the underlying credit survives before T, and nothing after the credit defaults. Calculate the expected value and the variance of the present value for this contract.
   (c) (5 points) What is the relationship between the two present values above?

4. (20 points) Suppose that we have a one-period default swap over time interval $[0, \Delta t]$. The premium P is paid at the beginning of the period, and the loss is paid at the time when default occurs, $1 - R$, where R is the recovery rate. We assume that the density function for the survival time is exponentially distributed, $f(t) = he^{-ht}$, and that we have a constant interest rate r over the time interval.
   (a) (5 points) Calculate the expected present value of loss leg for the one period default swap;

(b) (10 points) If default occurs at time t, and $0 \leq t \leq \Delta t$, then default protection buyer would receive a refund of $P(1 - \frac{t}{\Delta t})$ at the time of default from the protection seller. Calculate the expected present value of this refund.

(c) (5 points) What is the fair value of P for this one-period default swap?

5. (40 points) Download ISDA CDS Standard Model Excel addin from ISDA website for CDS Model, and the pricing sheet provided in the Course work.

   (a) (15 points) Use the standard model and a constant zero rate 5% to price a CDS deal with a maturity date of March 20, 2031, a running spread of 100 bps. The underlying credit has a flat CDS spread of 200 bps, and recovery rate of 40%. The valuation date is February 10, 2026. Present the value of the premium leg, loss leg, premium accrual, current MTM value, and a par credit default swap spread.

   (b) (5 points) If the protection buyer of the above trade would like to close the above position, would should the buyer do?

   (c) (20 points) Use the formulas provided in the lecture note, and use Python or Excel for the implementation of the formulas to check the numbers, and make comparison with the numbers from the Excel addin.